Parkinson's Disease

Parkinson's disease is a progressive neurological disorder that affects movement control. It is one of the most common neurodegenerative diseases worldwide, primarily affecting the elderly population. Early diagnosis and treatment are critical for managing the symptoms and slowing down disease progression.

This Parkinson's dataset, available on Kaggle, provides a rich collection of clinical and acoustic voice measurements to help develop AI models for early detection, progression monitoring, and treatment optimization.

This dataset is an essential resource for researchers and healthcare professionals focusing on neurological disorders. It includes detailed voice recordings and acoustic features that have been linked to Parkinson's disease symptoms.

The dataset provides valuable insights for developing machine learning models aimed at early diagnosis, monitoring disease progression, and predicting treatment outcomes.

Okomentoval(a): [.1]: How stable and what quality does the voice have it the dataset about

Okomentoval(a): [.2R1]: disrupts muscle control → voice is unstable, weaker, less regular

Patient Information

Name (Patient ID): Unique identifier for each patient.

Status: Indicates whether the individual is diagnosed with Parkinson's disease (1 for positive, 0 for negative).

Acoustic Features

MDVP (Hz): Average vocal fundamental frequency.

MDVP (Hz): Maximum vocal fundamental frequency.

MDVP (Hz): Minimum vocal fundamental frequency.

MDVP (%), MDVP (Abs), MDVP, MDVP, Jitter:DDP: Various measures of variations in frequency (jitter).

MDVP, MDVP(dB), Shimmer, Shimmer, Shimmer:DDA: Various measures of amplitude variation (shimmer).

NHR (Noise-to-Harmonics Ratio): Measures the proportion of noise to tonal components in the voice.

HNR (Harmonics-to-Noise Ratio): Quantifies the ratio of harmonic sound to noise in the voice.

Additional Clinical Features

RPDE (Recurrence Period Density Entropy): A nonlinear dynamical analysis of voice.

DFA (Detrended Fluctuation Analysis): Measures the long-term signal correlation in the voice.

Spread1, Spread2: Measures of variation in voice frequency.

D2: Another nonlinear dynamical measure.

PPE (Pitch Period Entropy): Measures the regularity of the pitch period.

Okomentoval(a): [.3]: FREQUENCY

how "high" or "deep" the voice is

average, max, min frequency

Okomentoval(a): [.4R3]: differences between people:

is the voice monotonous?
is it less ranged?

Okomentoval(a): [.5]: JITTER (frequency fluctuation)
how much the frequency "jumps"

Okomentoval(a): [.6R5]: higher jitter = worse voice control

Okomentoval(a): [.7]: SHIMMER (volume fluctuation)
how the volume fluctuates
we look for an unstable voice:

sometimes loud
sometimes soft

Okomentoval(a): [.8]: NOISE (noise in voice)
NHR=how much noise is there

Okomentoval(a): [.9]: HNR=how much "clean" voice is there

Okomentoval(a): [.10]: signal complexity
high value = chaotic voice

Okomentoval(a): [.11]: long-term patterns
shows how "organized" the signal is

Okomentoval(a): [.12]: frequency dispersion

Okomentoval(a): [.13]: system complexity

Okomentoval(a): [.14]: voice regularity
higher = more irregularity

Data Preprocessing

Data Cleaning: Ensure that no missing or inconsistent values exist in the dataset. Handle outliers in voice measurements, ensuring clean and accurate data for model input.

Normalization: Normalize features such as frequency and jitter measures to ensure consistency across samples before feeding them into machine learning models. Standardization can be applied to continuous variables like Fo, Fhi, and Flo.

Model Training

Frameworks: Leverage popular machine learning frameworks such as TensorFlow, PyTorch, or scikit-learn to train models on this dataset.

Model Selection: Algorithms like Support Vector Machines (SVM), Decision Trees, Random Forests, or Deep Neural Networks can be explored to classify patients as having Parkinson's disease or not, based on their voice metrics.

Evaluation: Evaluate model performance using metrics like accuracy, precision, recall, and F1-score to ensure the effectiveness of models in classifying Parkinson's disease.

Deployment

Clinical Decision Support: Integrate the trained model into clinical tools to assist healthcare providers in diagnosing Parkinson's disease early based on voice recordings.

Testing and Feedback: Continuously test the tool in clinical settings to ensure its reliability. Incorporate feedback from healthcare professionals to improve the model's accuracy and usability.

Potential Applications

Machine Learning Models: Develop algorithms for early diagnosis and monitoring of Parkinson's disease progression, as well as predicting patient outcomes based on vocal patterns.

Healthcare Insights: Provide clinicians with valuable insights for better understanding the vocal characteristics associated with Parkinson's disease.

Academic Research: Facilitate research into vocal biomarkers for Parkinson's disease, the development of non-invasive diagnostic tools, and personalized treatment plans.

Conclusion

The Parkinson's Disease dataset offers an invaluable resource for the medical and research community. With its rich collection of acoustic measurements, it provides a robust foundation for the development of AI-based solutions to improve Parkinson's disease diagnosis and treatment.

What to do with it...

firstly

differences between healthy vs Parkinson

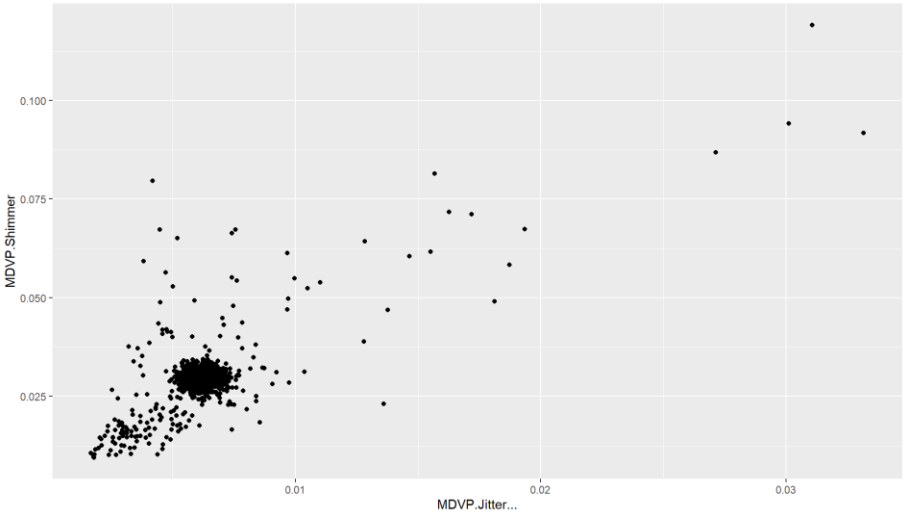
we should watch when jitter ↑, shimmer ↑, noise ↑, entropy ↑

relationships between variables jitter vs shimmer and noise vs jitter

voice stability: How stable is the voice?

Very basic analysis

```
ggplot(parkins, aes(x = MDVP.Jitter..., y = MDVP.Shimmer)) +
geom_point()
```



The scatter plot reveals a positive relationship between jitter and shimmer, indicating that increased frequency instability is associated with increased amplitude variability in vocal signals.

```
> summary(parkins$`MDVP.Jitter...`)
```

Min. 1st Qu. Median Mean 3rd Qu. Max.

0.001680 0.005769 0.006188 0.006221 0.006557 0.033160

```
> summary(parkins$`MDVP.Shimmer`)
```

Min. 1st Qu. Median Mean 3rd Qu. Max.

0.00954 0.02799 0.02958 0.02967 0.03101 0.11908

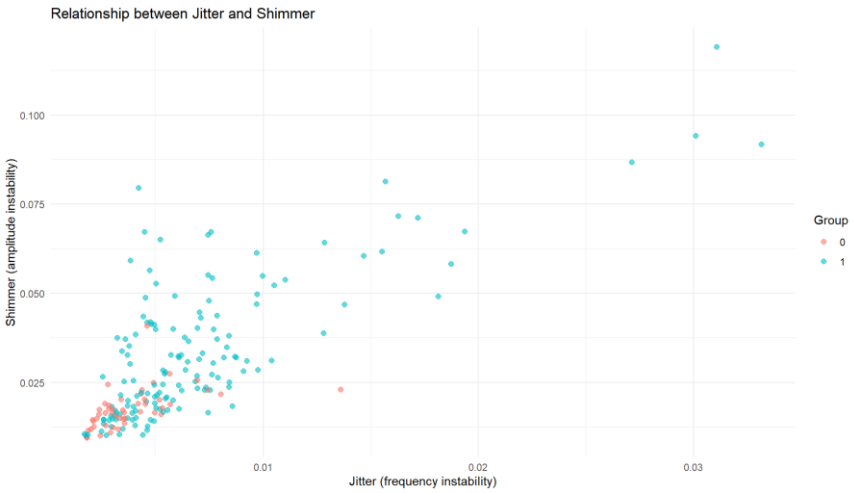
Which was odd because status should be just 0 or 1

But from 195 row it was sort of random floating point so now working with 0-195 rows

```
table(parkins_clean$status)
```

48 Healthy

147 Parkinson



Healthy individuals show more consistent vocal characteristics, while Parkinson patients demonstrate increased dispersion and extreme values.

Basic question

How do acoustic voice features differ between healthy individuals and Parkinson's patients?

Main question of the project

Can we combine acoustic features into a single measure of vocal instability that distinguishes Parkinson's disease?

Added value

Can vocal instability be modeled as a multidimensional phenomenon, and does this structure reveal patterns specific to Parkinson's disease?

What chat suggested - hard level

To what extent can combined acoustic features serve as a reliable biomarker for neurological impairment, and what is their explanatory power compared to individual features?

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